

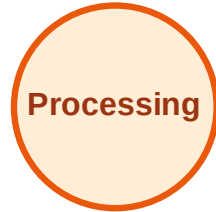
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



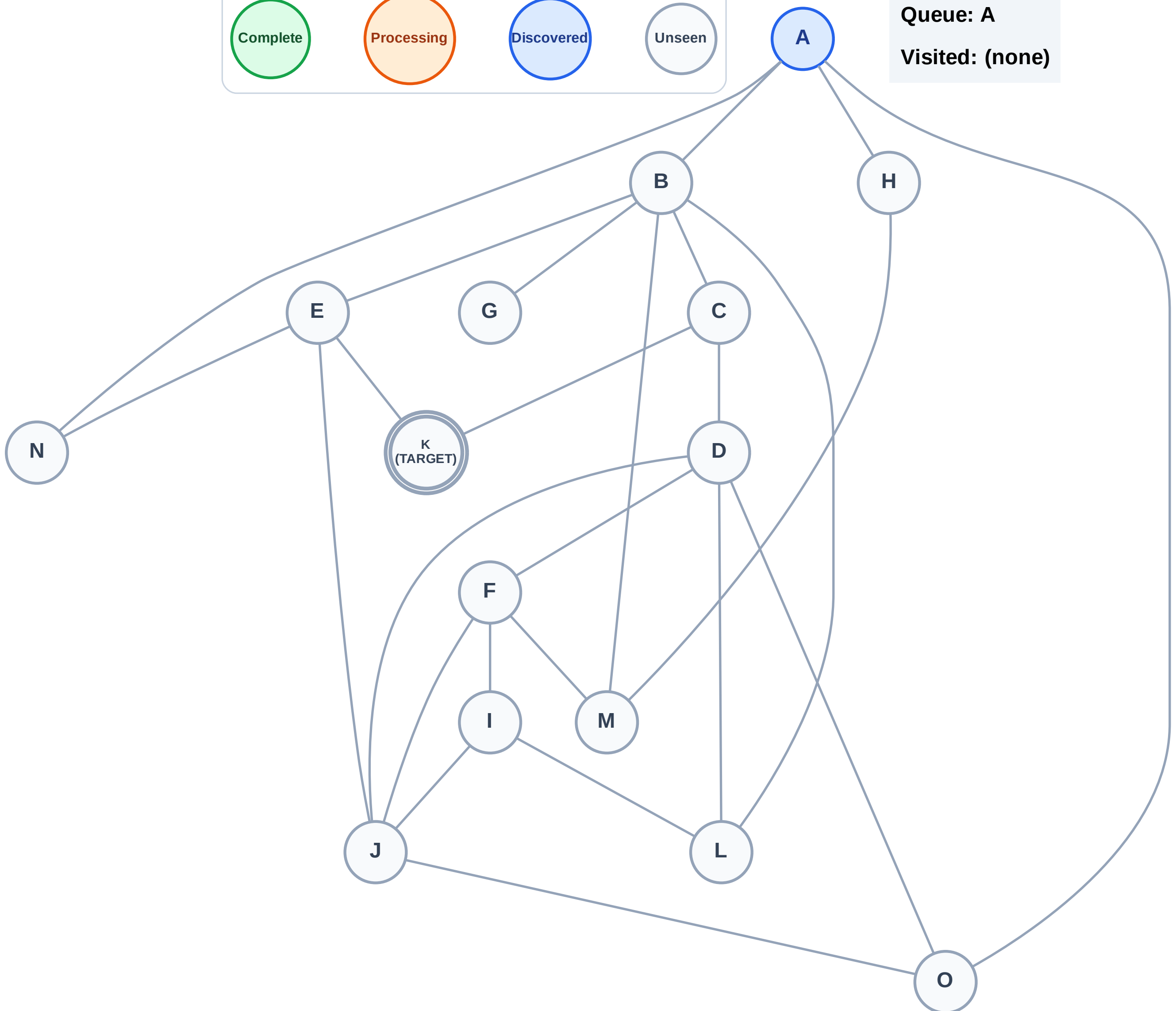
Discovered



Unseen

Queue: A

Visited: (none)



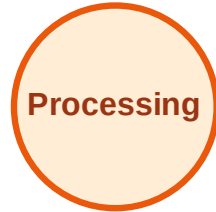
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



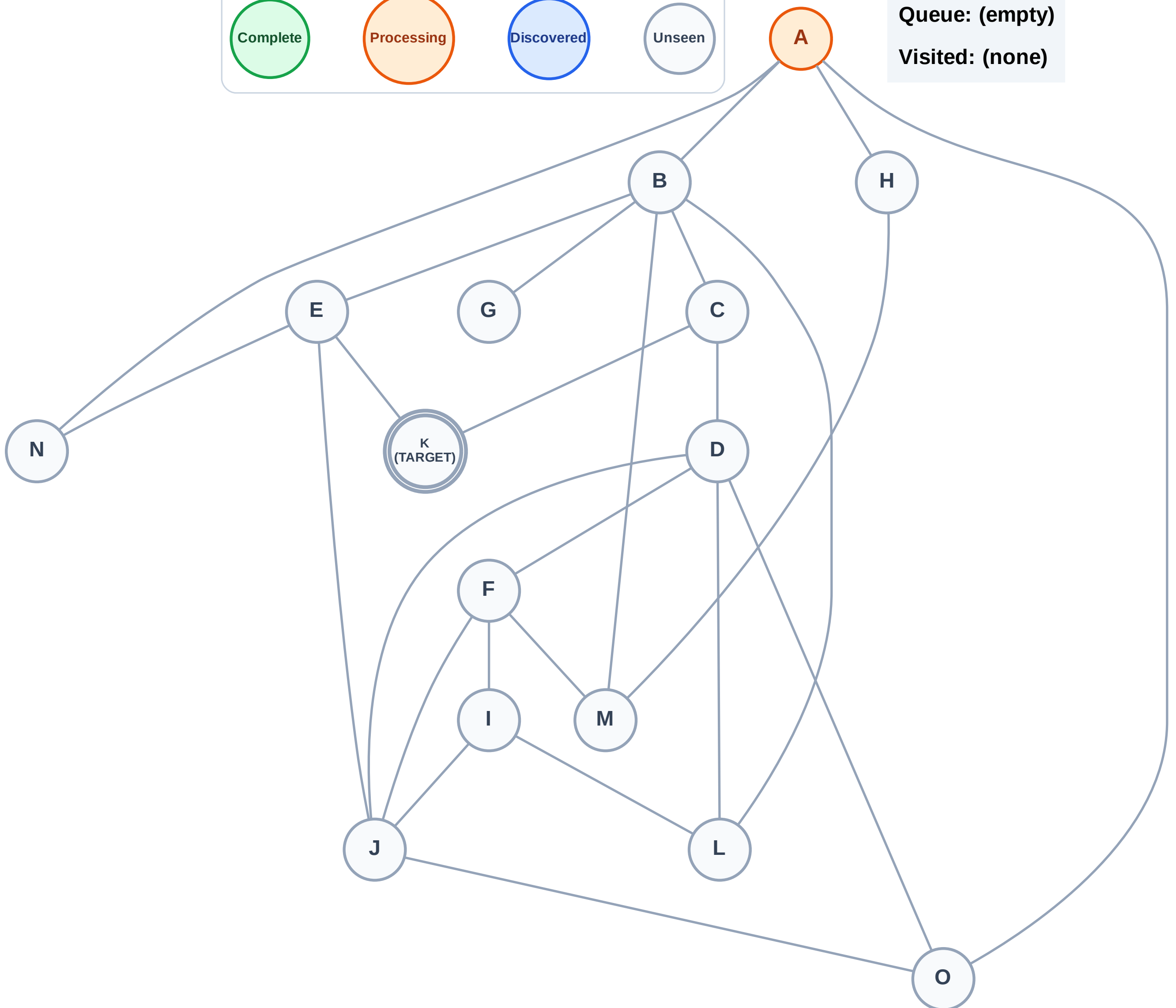
Discovered



Unseen

Queue: (empty)

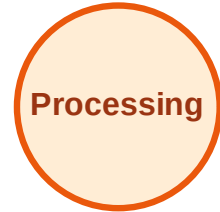
Visited: (none)



BFS Traversal

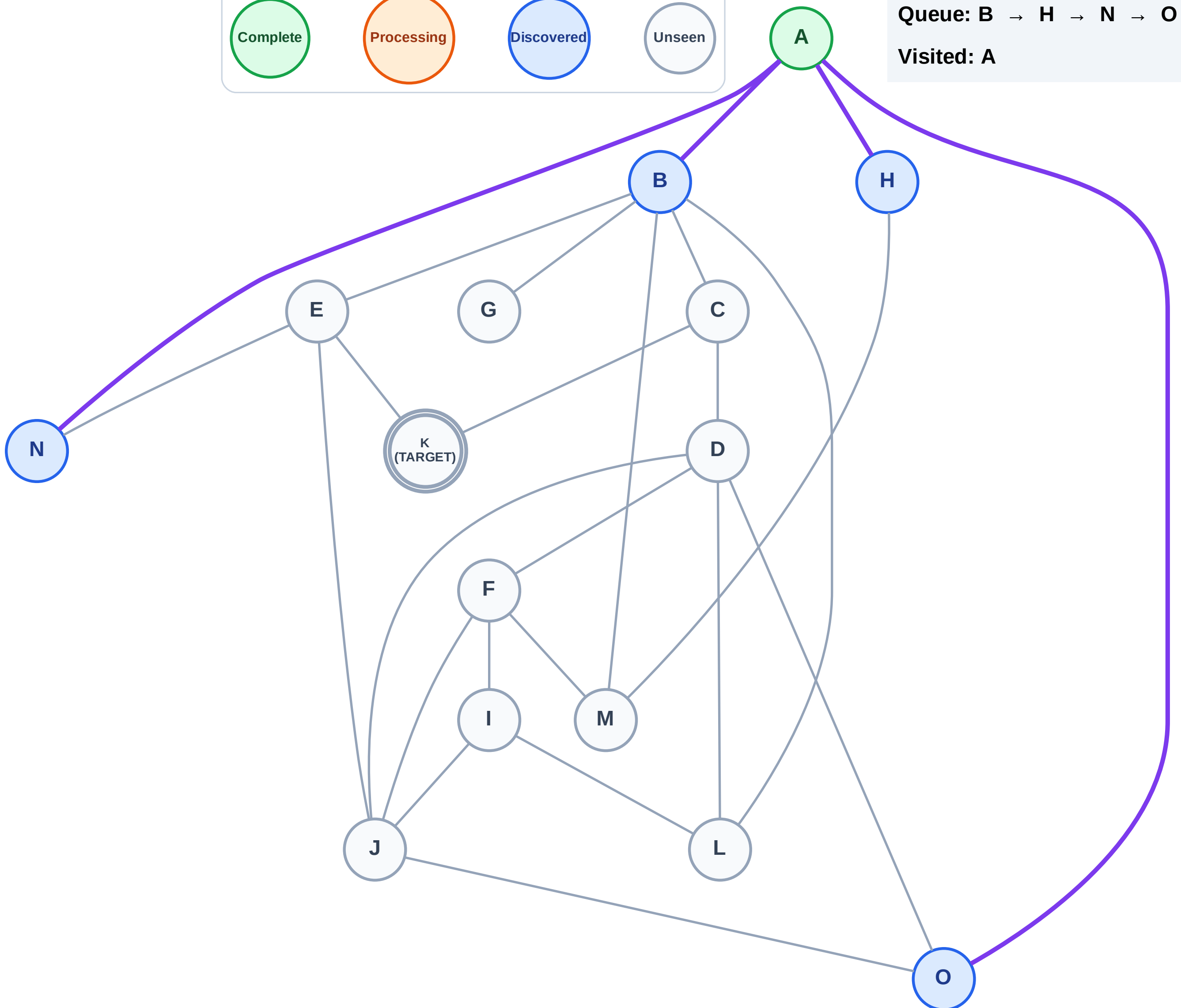
Step 3: Discover B, H, N, O from A

Legend



Queue: B → H → N → O

Visited: A



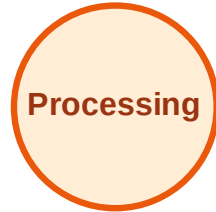
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



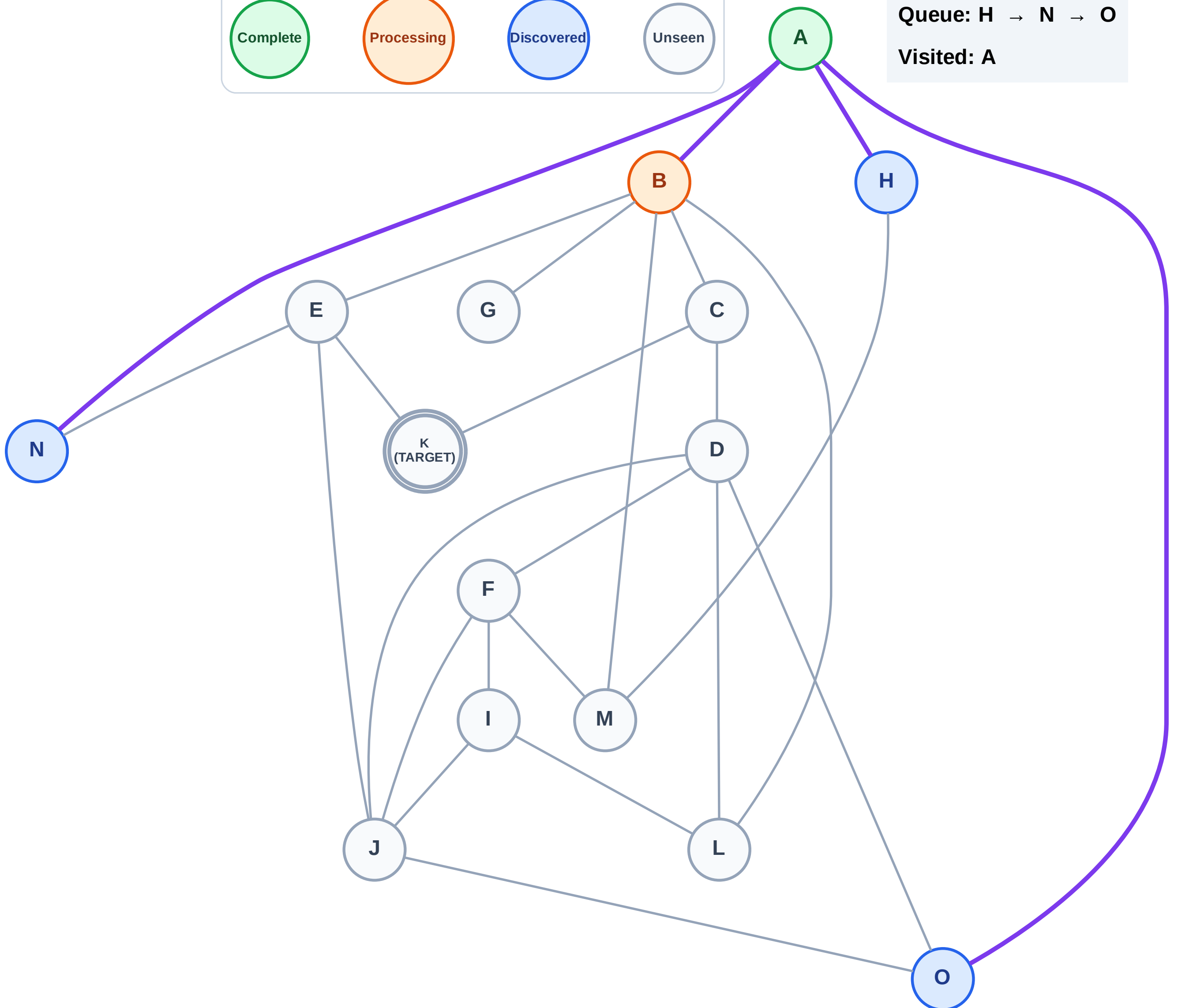
Discovered



Unseen

Queue: H → N → O

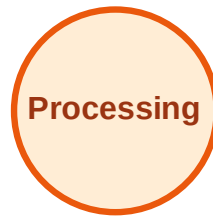
Visited: A



BFS Traversal

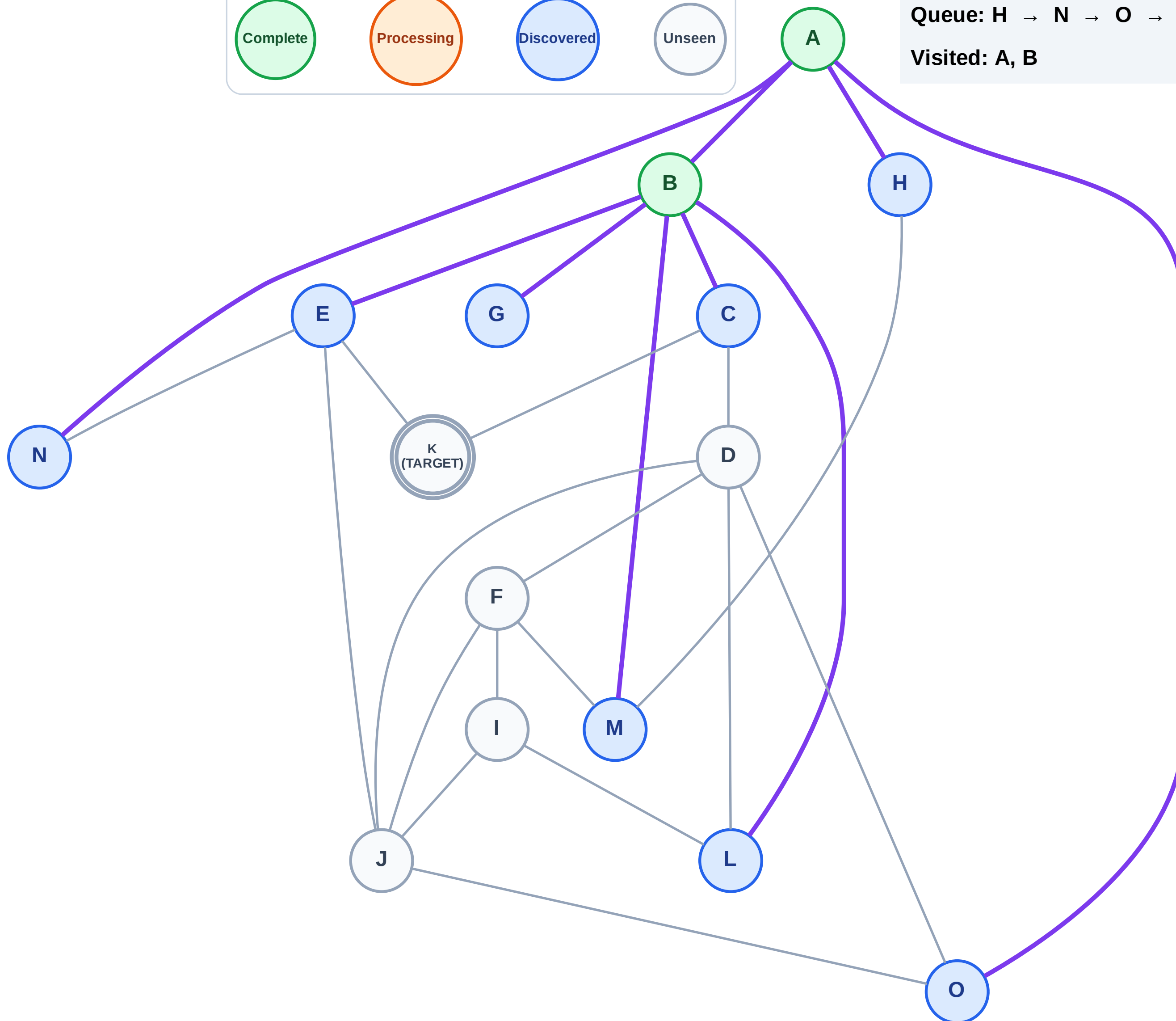
Step 5: Discover C, E, G, L, M from B

Legend



Queue: H → N → O → C → E → G → L → M

Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

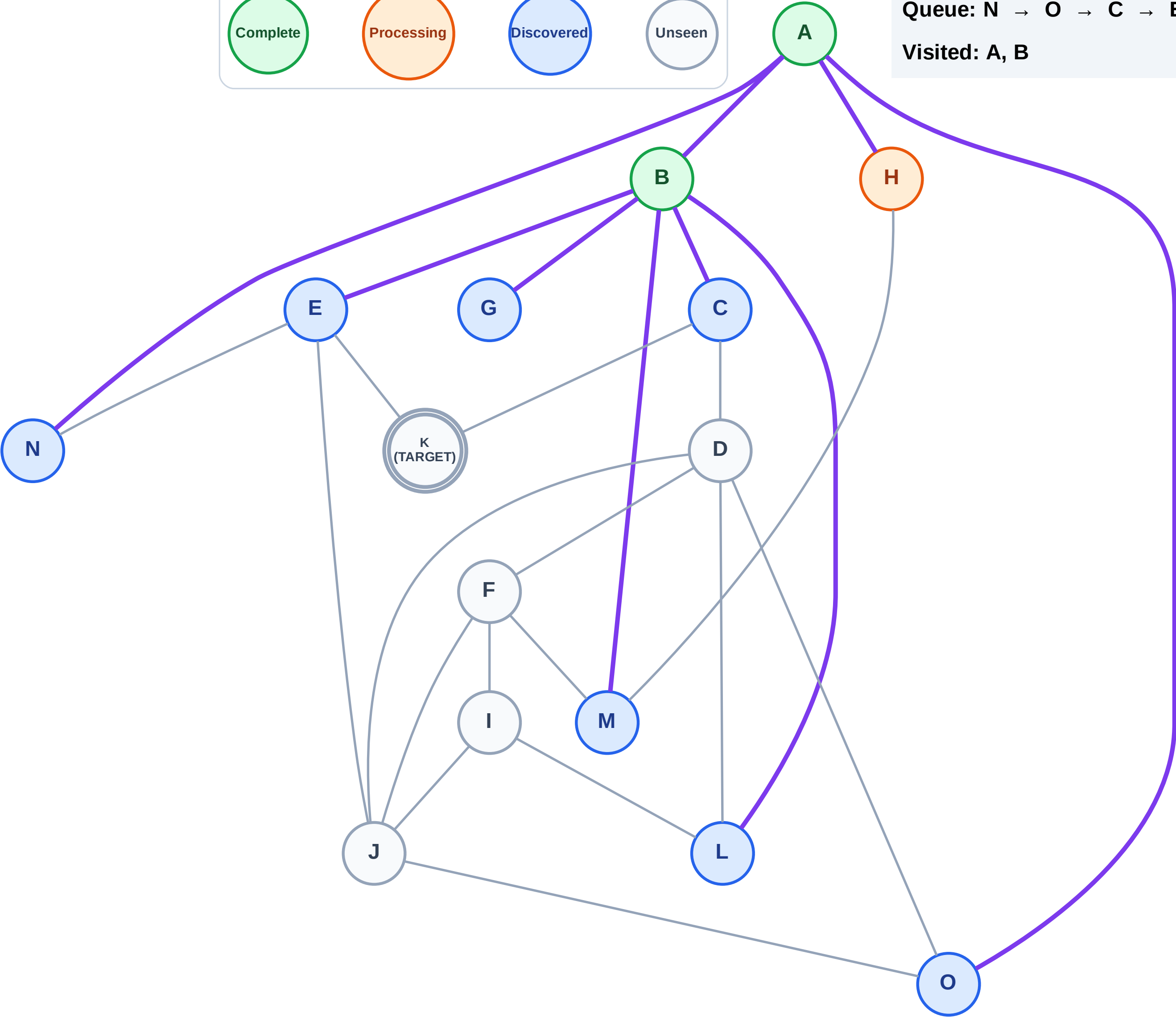
Processing

Discovered

Unseen

Queue: N → O → C → E → G → L → M

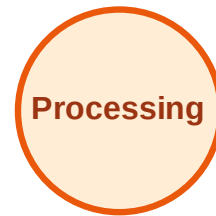
Visited: A, B



BFS Traversal

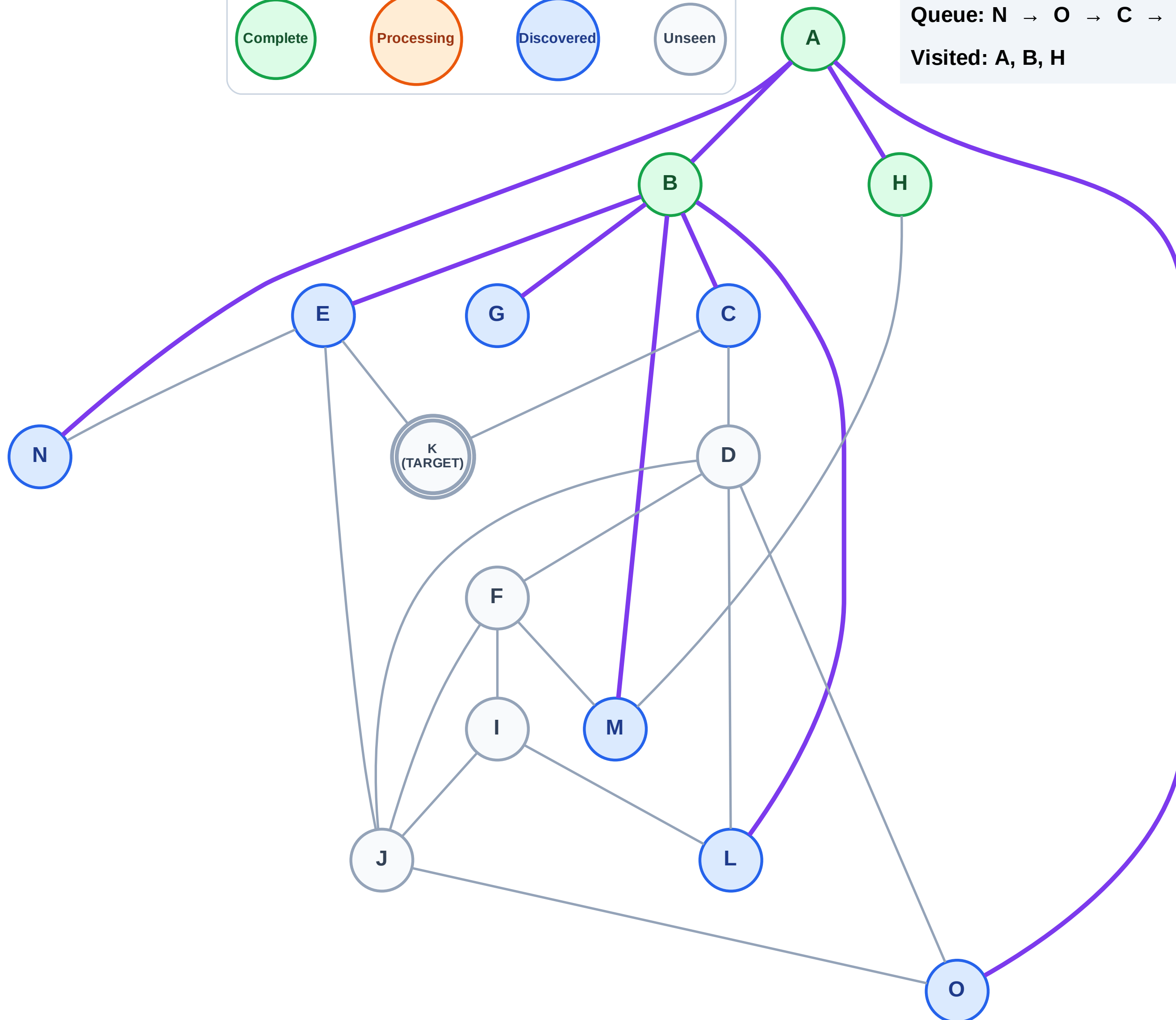
Step 7: No new neighbors from H

Legend



Queue: N → O → C → E → G → L → M

Visited: A, B, H



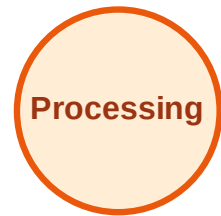
BFS Traversal

Step 8: Process node N

Legend



Complete



Processing



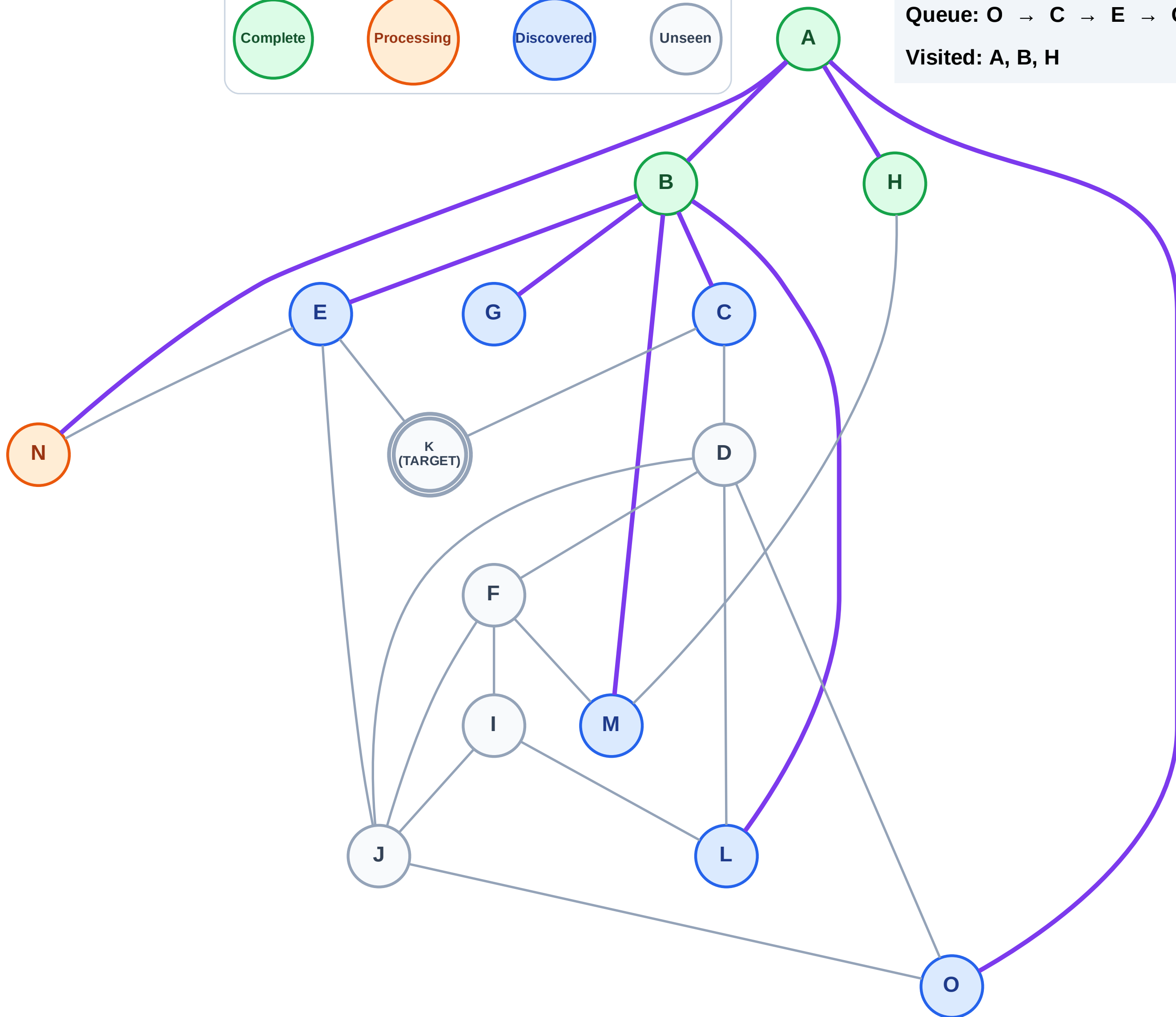
Discovered



Unseen

Queue: O → C → E → G → L → M

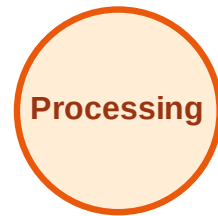
Visited: A, B, H



BFS Traversal

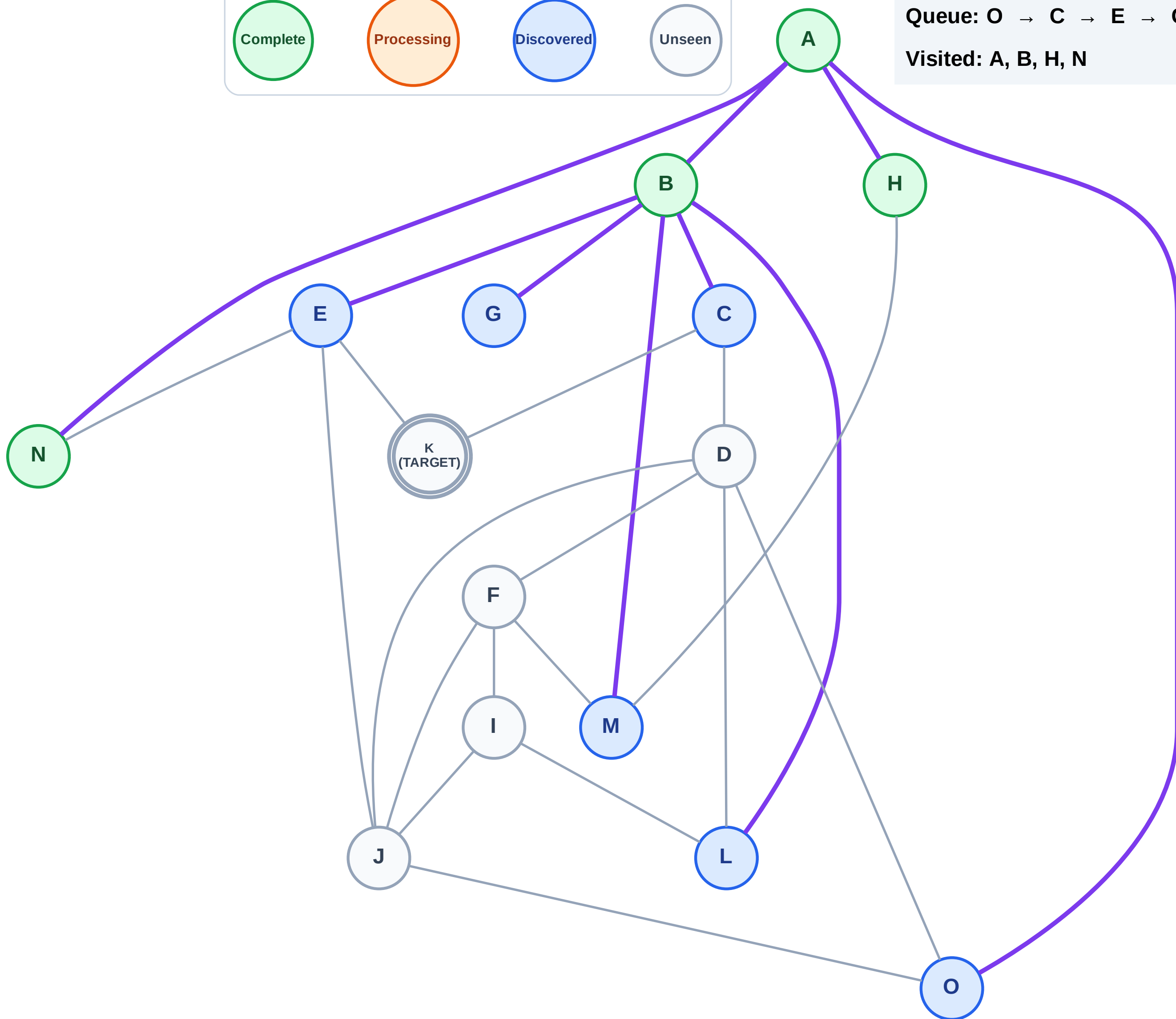
Step 9: No new neighbors from N

Legend



Queue: O → C → E → G → L → M

Visited: A, B, H, N



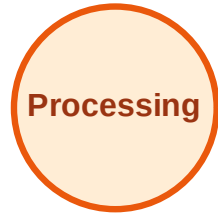
BFS Traversal

Step 10: Process node O

Legend



Complete



Processing



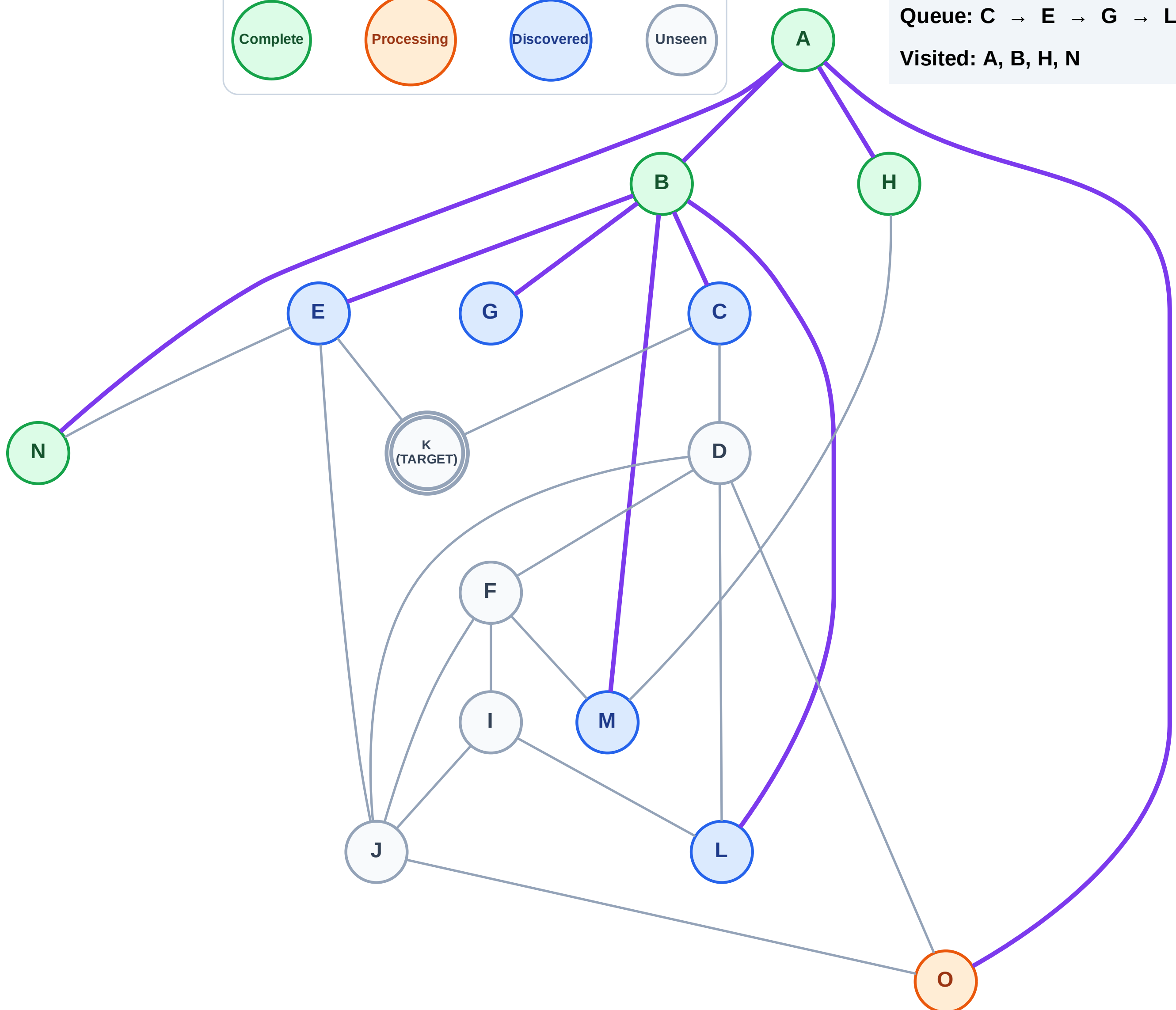
Discovered



Unseen

Queue: C → E → G → L → M

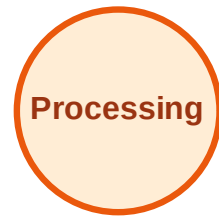
Visited: A, B, H, N



BFS Traversal

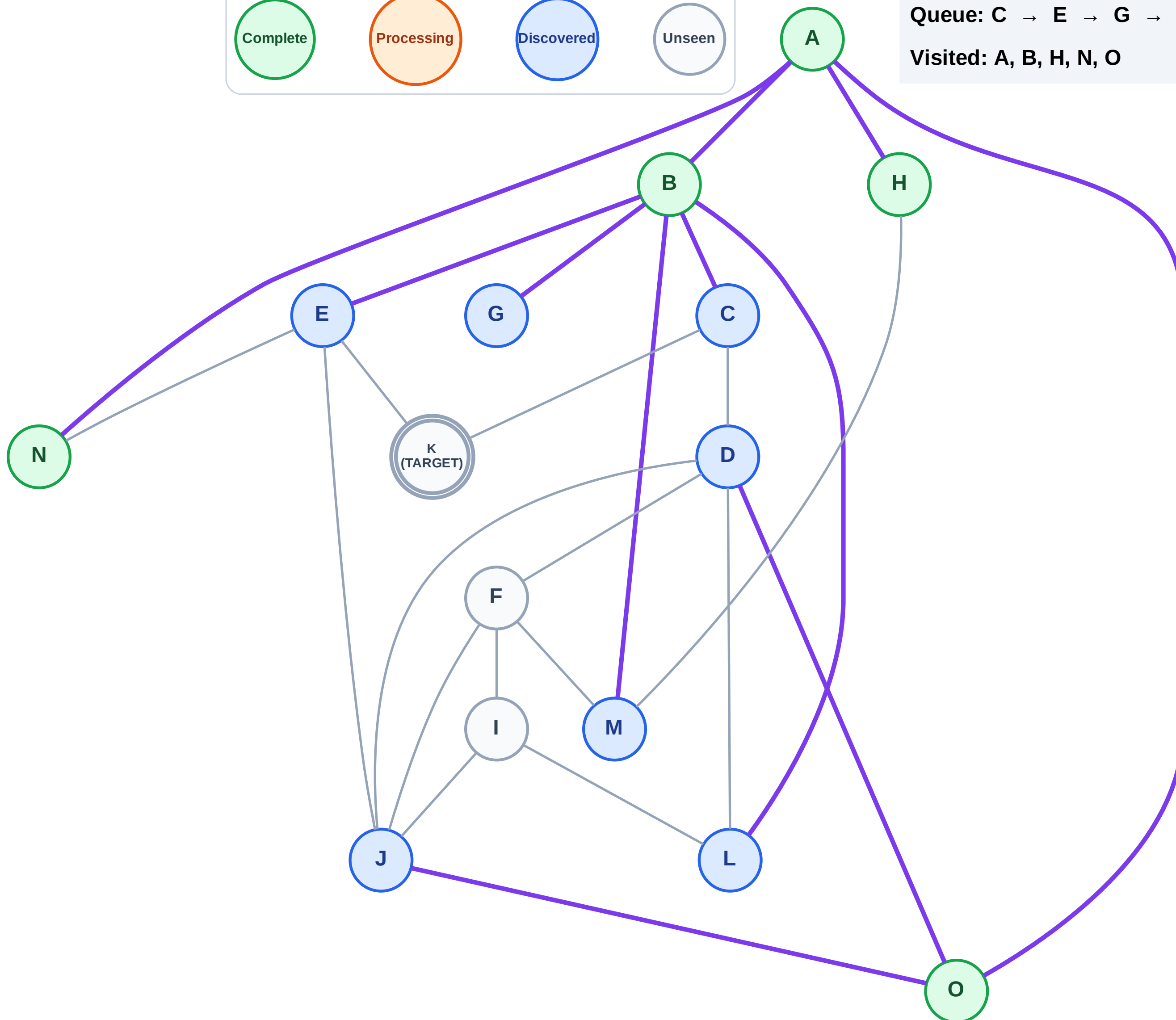
Step 11: Discover D, J from O

Legend



Queue: C → E → G → L → M → D → J

Visited: A, B, H, N, O



BFS Traversal

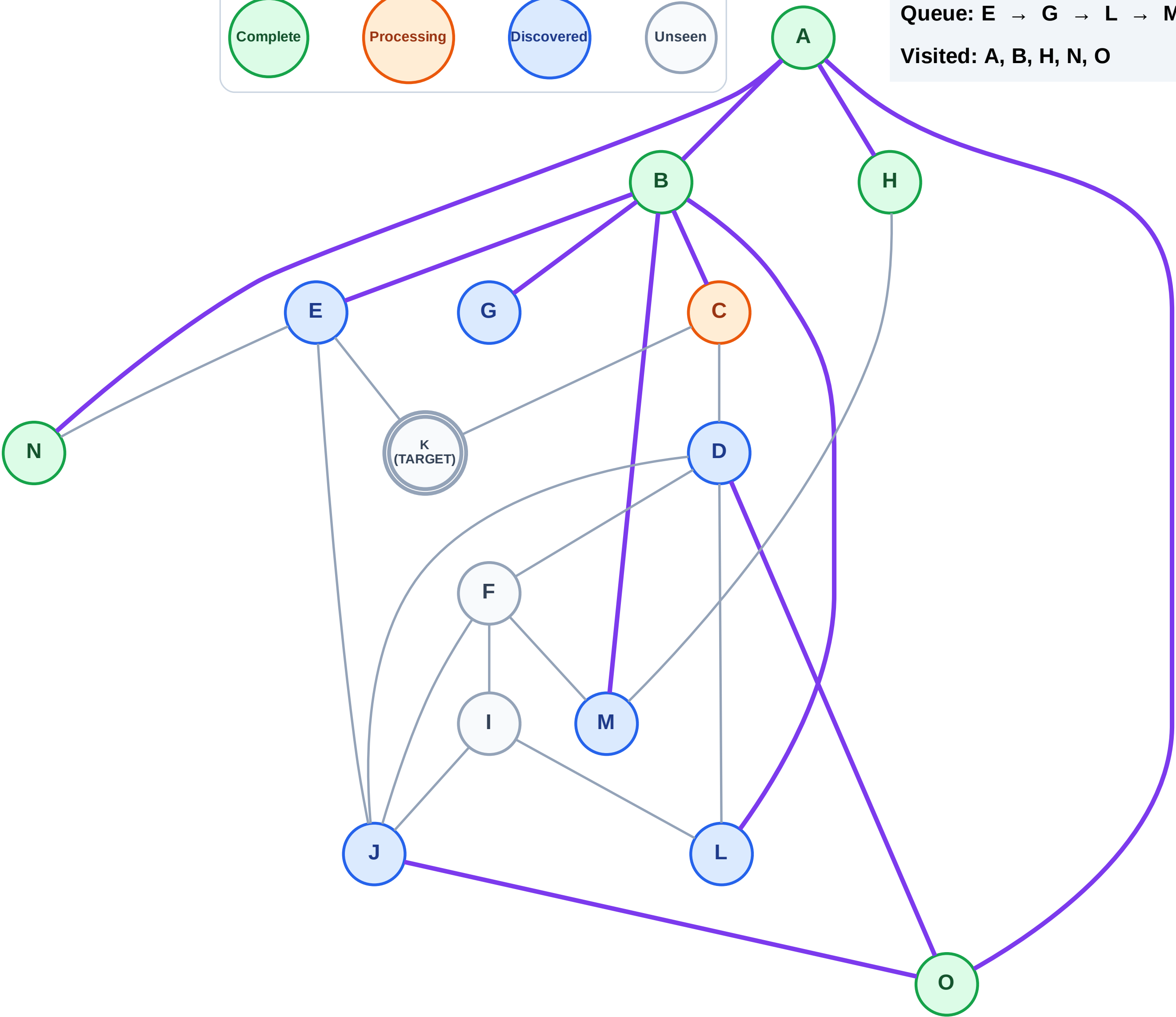
Step 12: Process node C

Legend



Queue: E → G → L → M → D → J

Visited: A, B, H, N, O



BFS Traversal

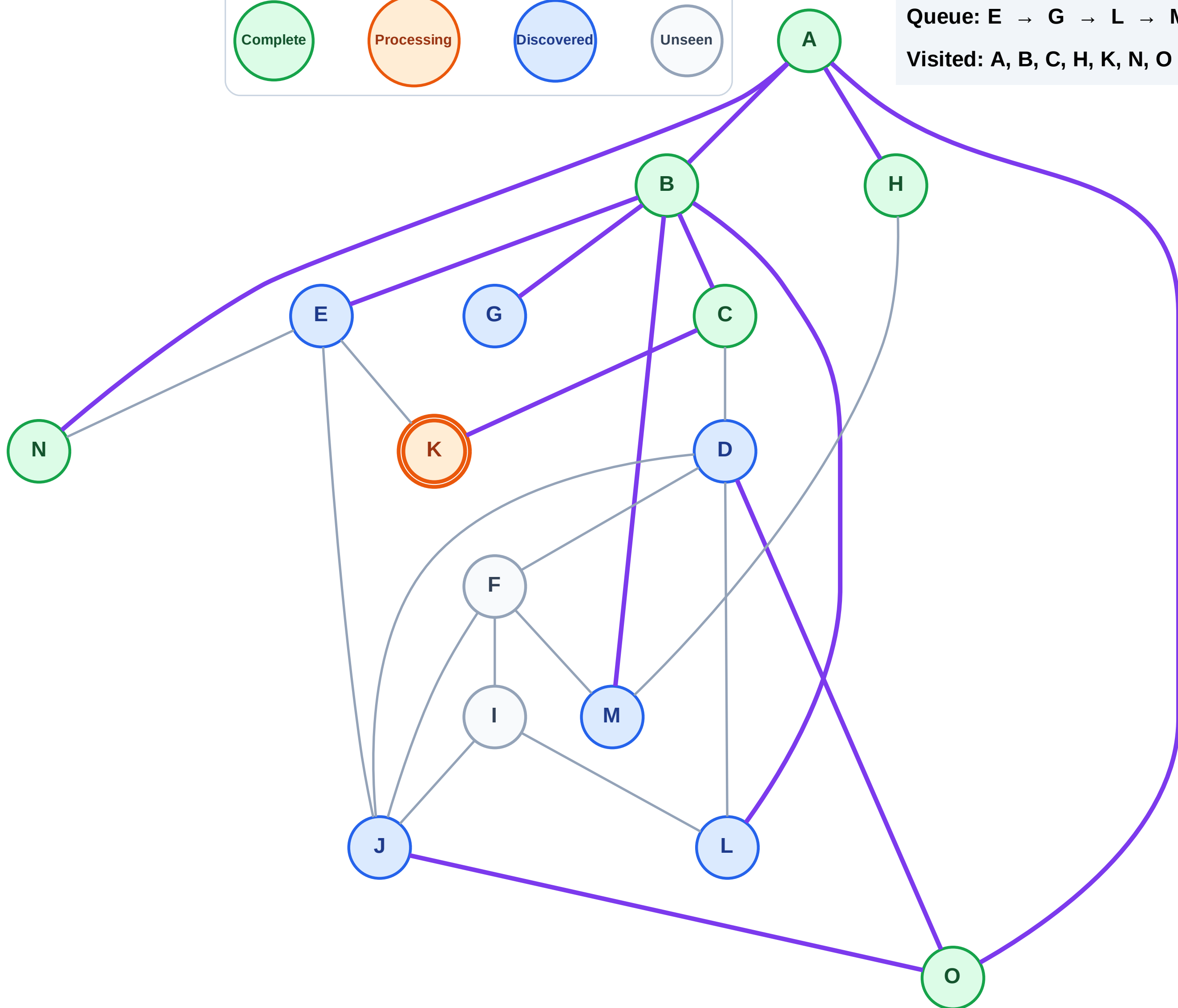
Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Queue: E → G → L → M → D → J

Visited: A, B, C, H, K, N, O



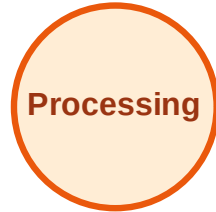
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



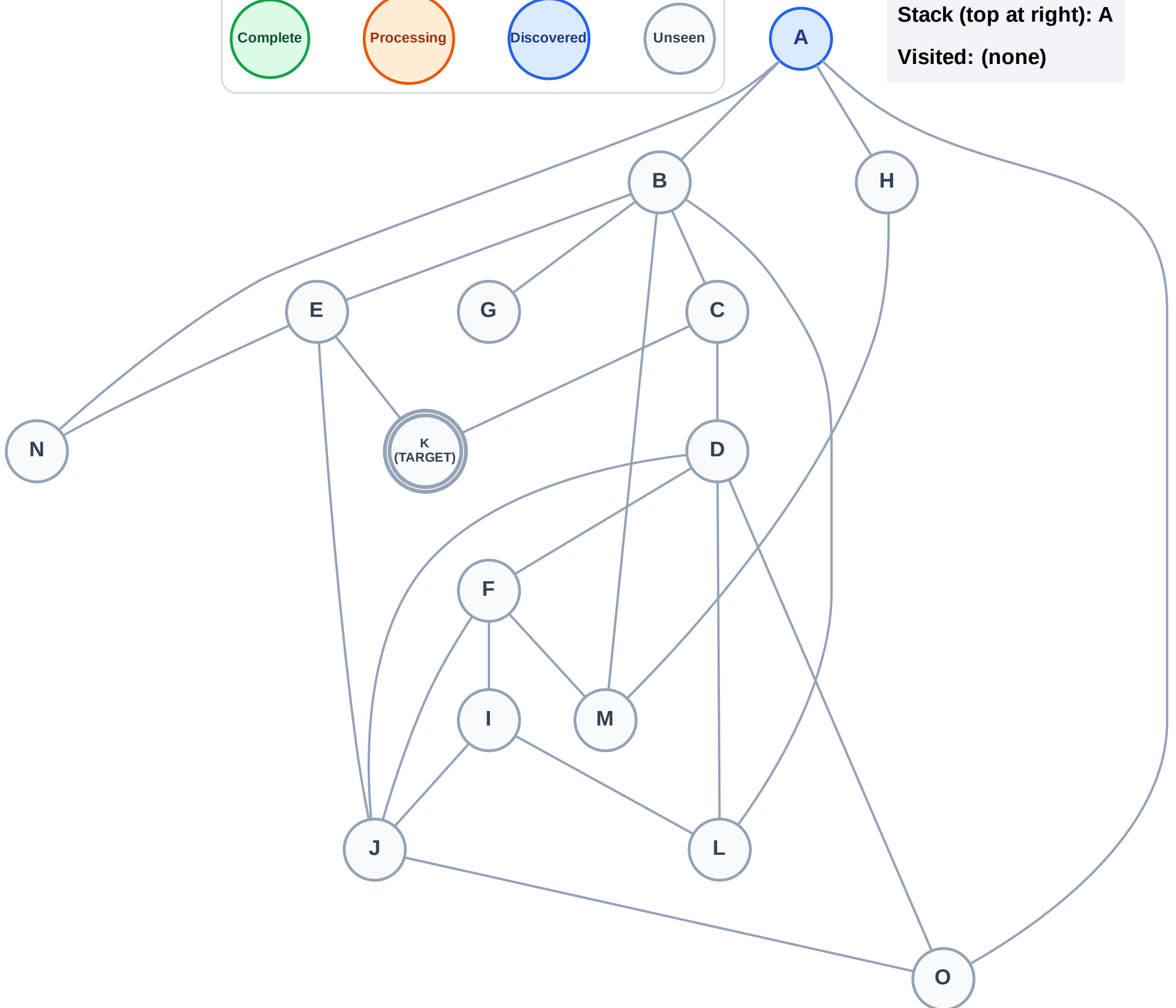
Discovered



Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

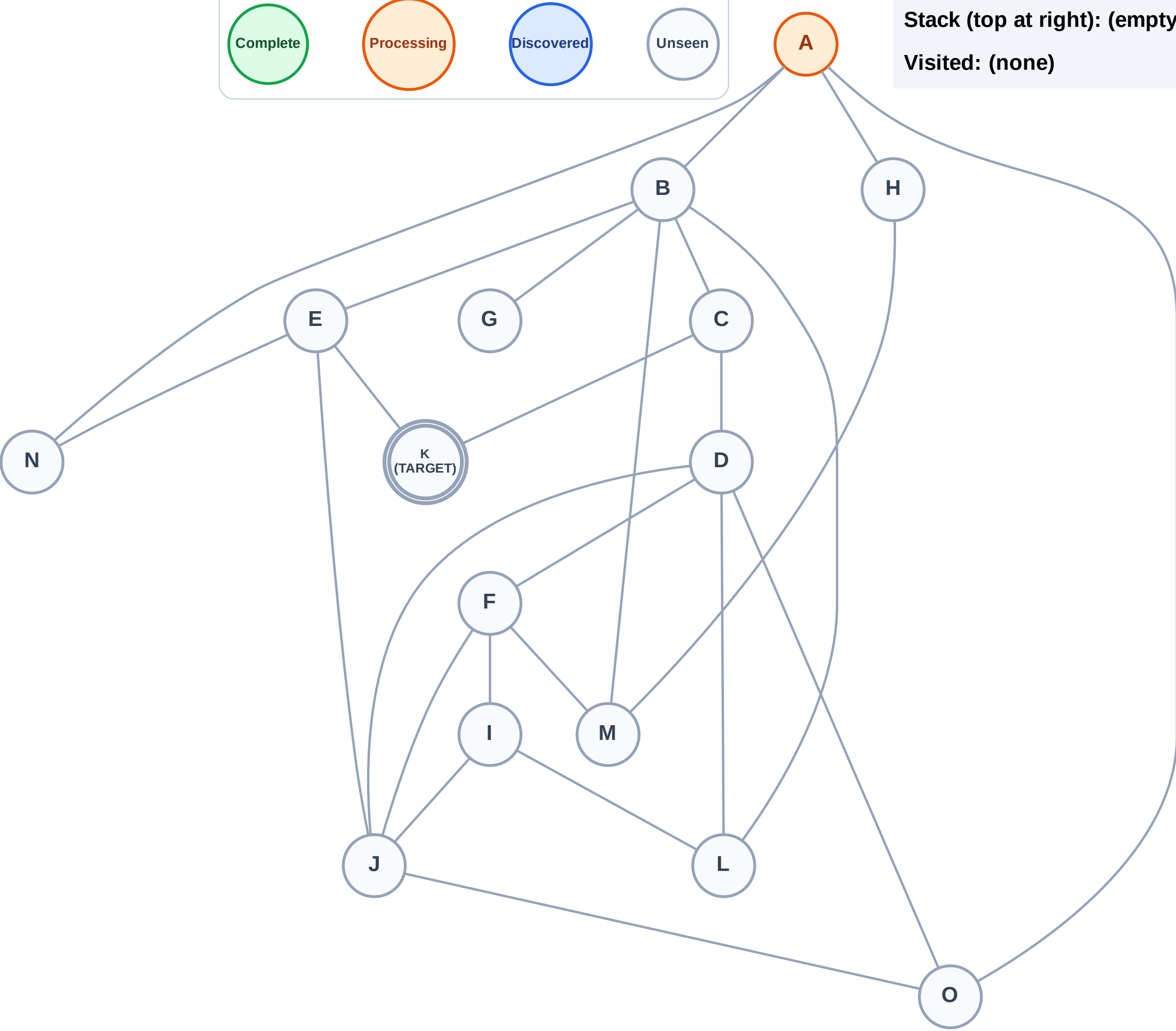
Complete

Processing

Discovered

Unseen

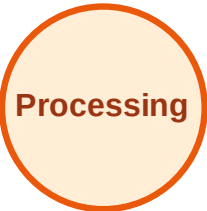
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

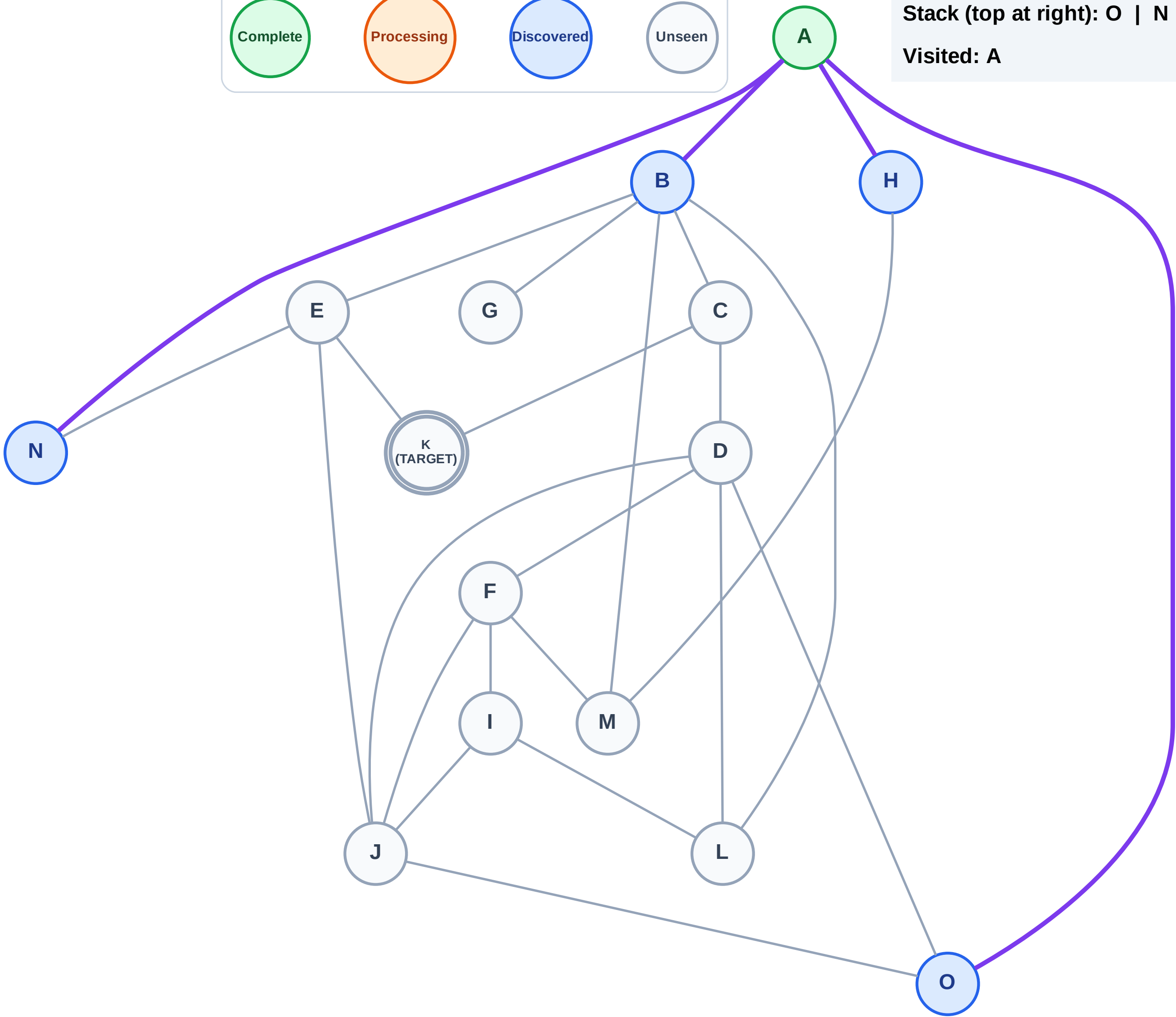
Step 3: Discover O, N, H, B from A

Legend



Stack (top at right): O | N | H | B

Visited: A



DFS Traversal

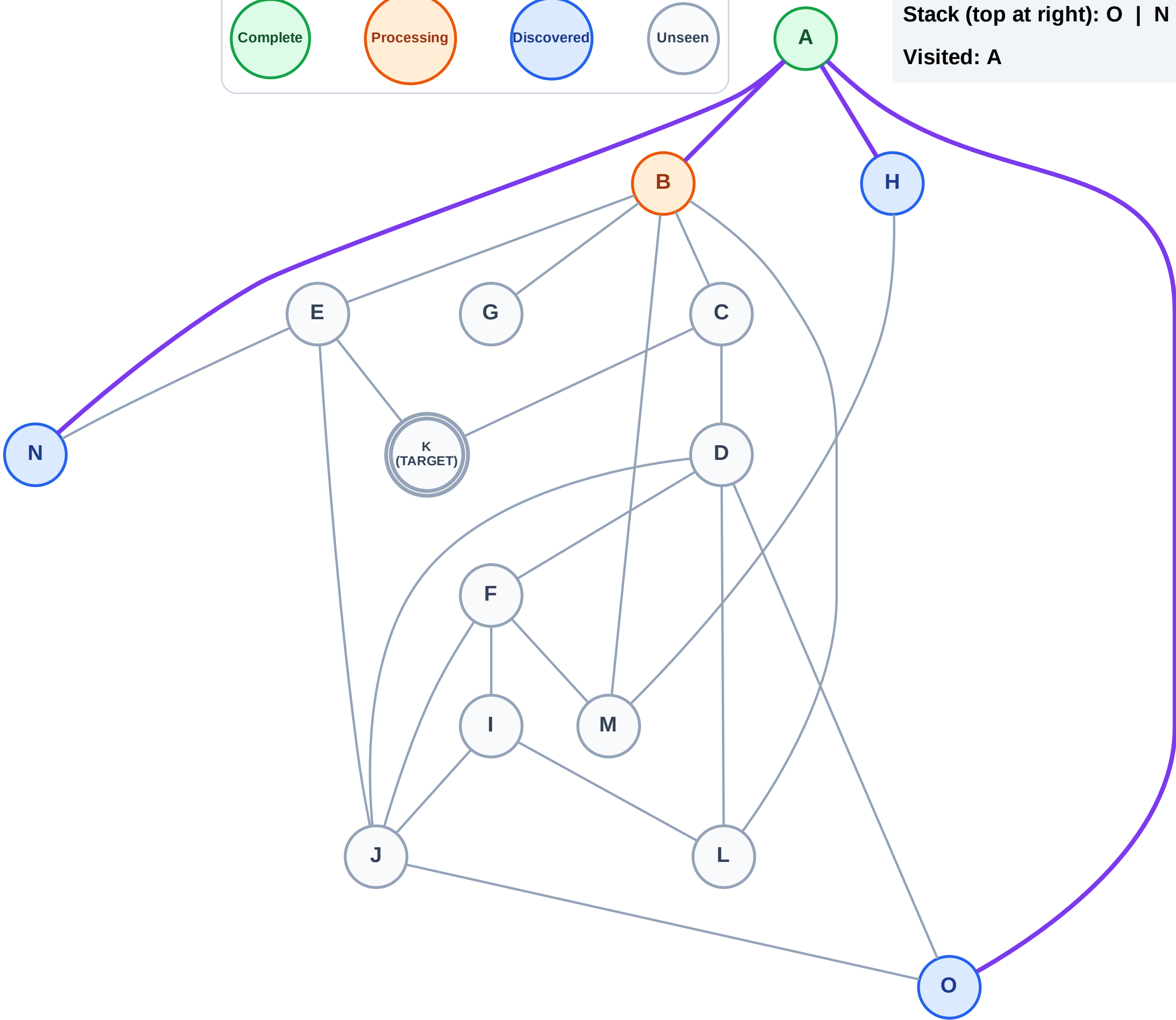
Step 4: Process node B

Legend



Stack (top at right): O | N | H

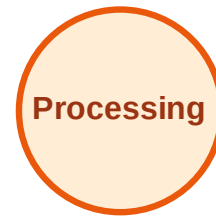
Visited: A



DFS Traversal

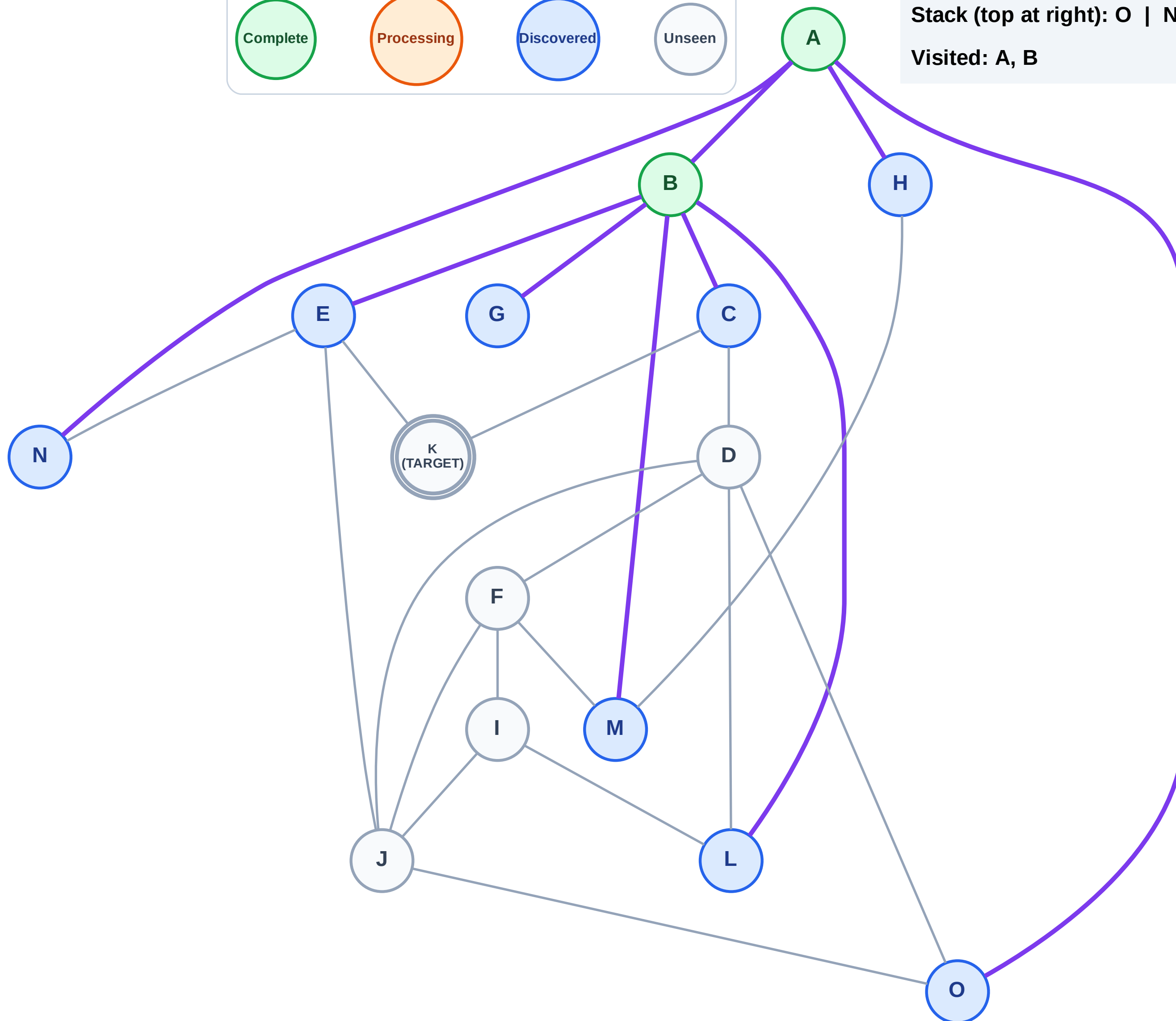
Step 5: Discover M, L, G, E, C from B

Legend



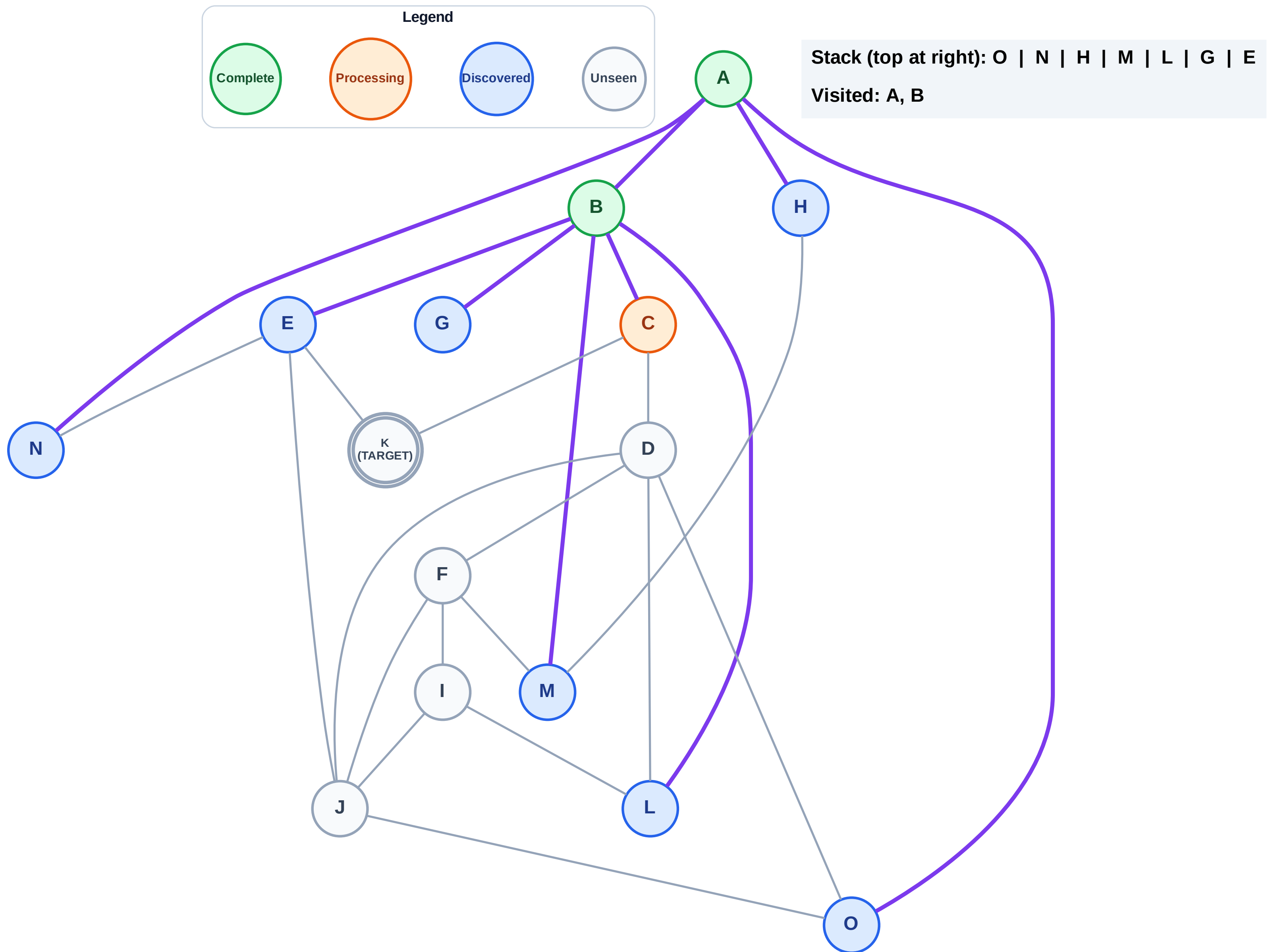
Stack (top at right): O | N | H | M | L | G | E | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | N | H | M | L | G | E

Visited: A, B, C, K

